Name:Tharunya BS Register Number: 250701786
Experiment Name: Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

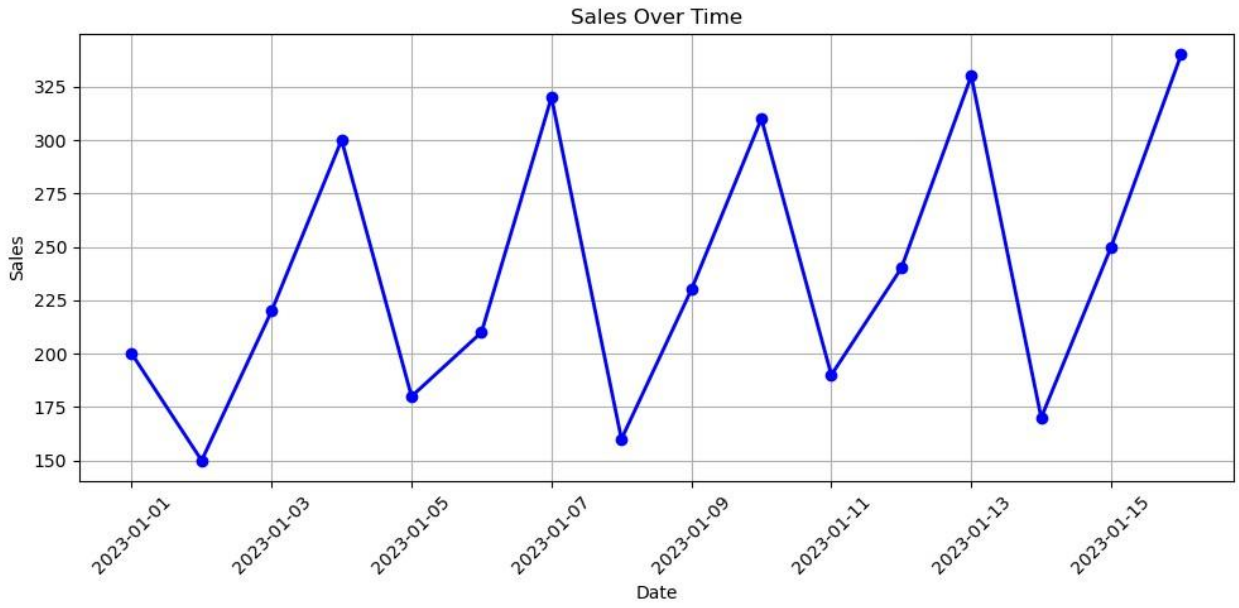
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# ------------------------------------------------
# 1. CREATE DATASET
#------------------------------------------------
data = {     "Date":[
        "01-01-2023", "02-01-2023", "03-01-2023", "04-01-2023",
        "05-01-2023", "06-01-2023", "07-01-2023", "08-01-2023",
        "09-01-2023", "10-01-2023", "11-01-2023", "12-01-2023",
        "13-01-2023",  "14-01-2023",  "15-01-2023",  "16-01-2023"
],
    "Product": [
        "Product A", "Product B", "Product A", "Product C",
        "Product B", "Product A", "Product C", "Product B",
        "Product A", "Product C", "Product B", "Product A",
        "Product C", "Product B", "Product A", "Product C"
    ],
    "Sales": [
        200, 150, 220, 300, 180, 210, 320, 160,
        230, 310, 190, 240, 330, 170, 250, 340
    ],
    "Quantity": [
        4, 3, 5, 6, 4, 5, 7, 3,
        6, 7, 4, 6, 8, 3, 7, 8
    ],
    "Region": [
        "North", "South", "North", "East",
        "West", "North", "East", "South",
        "North", "East", "West", "North",
        "East", "South", "North", "East"
    ]
}
# Create DataFrame df =
pd.DataFrame(data) #
Convert Date column
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)
# ------------------------------------------------
# 2. DISPLAY DATA
# ------------------------------------------------
print("DATASET:") print(df.head()) print("\n")
# ------------------------------------------------
# 3. CHECK MISSING VALUES
```

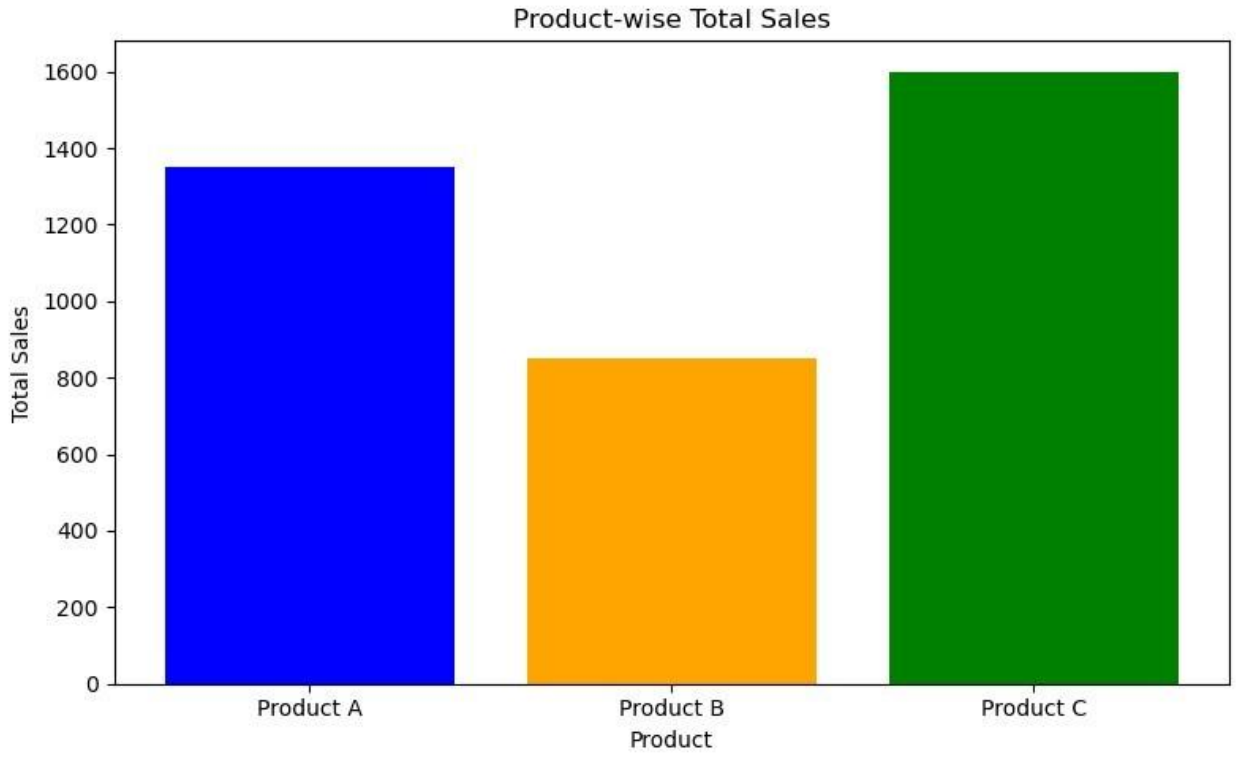
```
# ------------------------------------------------
print("MISSING VALUES:") print(df.isnull().sum())
print("\n")
# ------------------------------------------------
# 4. STATISTICAL SUMMARY
# ------------------------------------------------print("STATISTICAL
SUMMARY:")
print(df[["Sales", "Quantity"]].describe())
print("\n")
# ------------------------------------------------
# 5. PRODUCT-WISE SALES AND QUANTITY
# ------------------------------------------------product_summary
= df.groupby("Product")[["Sales",
"Quantity"]].sum().reset_index()
print("PRODUCT-WISE SALES AND QUANTITY:")
print(product_summary) print("\n")
# ------------------------------------------------
# 6. LINE GRAPH - SALES OVER DATE
# ------------------------------------------------
plt.figure(figsize=(10, 5)) plt.plot(
df["Date"],     df["Sales"],     marker="o",
color="blue",     linewidth=2
)
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
# ------------------------------------------------
# 7. BAR GRAPH - PRODUCT-WISE SALES
# ------------------------------------------------
plt.figure(figsize=(8, 5)) plt.bar(
    product_summary["Product"],
product_summary["Sales"],
    color=["blue", "orange", "green"]
)
plt.title("Product-wise Total Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.tight_layout()
```

```
plt.show()
# ------------------------------------------------
# 8. CORRELATION MATRIX
# ------------------------------------------------
correlation = df[["Sales", "Quantity"]].corr()
print("CORRELATION MATRIX:") print(correlation)
print("\n")
# ------------------------------------------------
# 9. CORRELATION HEATMAP
# ------------------------------------------------
plt.figure(figsize=(7, 5)) sns.heatmap(
correlation,     annot=True,     cmap="coolwarm",
linewidths=0.5,     fmt=".2f"
)
plt.title("Correlation Matrix")
plt.tight_layout() plt.show()
DATASET:
        Date    Product  Sales  Quantity Region
0 2023-01-01  Product A    200         4  North
1 2023-01-02  Product B    150         3  South
2 2023-01-03  Product A    220         5  North
3 2023-01-04  Product C    300         6   East
4 2023-01-05  Product B    180         4   West
MISSING VALUES:
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64


 STATISTICAL SUMMARY:
            Sales   Quantity
 count   16.000000  16.000000
 mean   237.500000   5.375000
 std     64.031242   1.746425min    150.000000   3.000000
                                  25%
187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
```

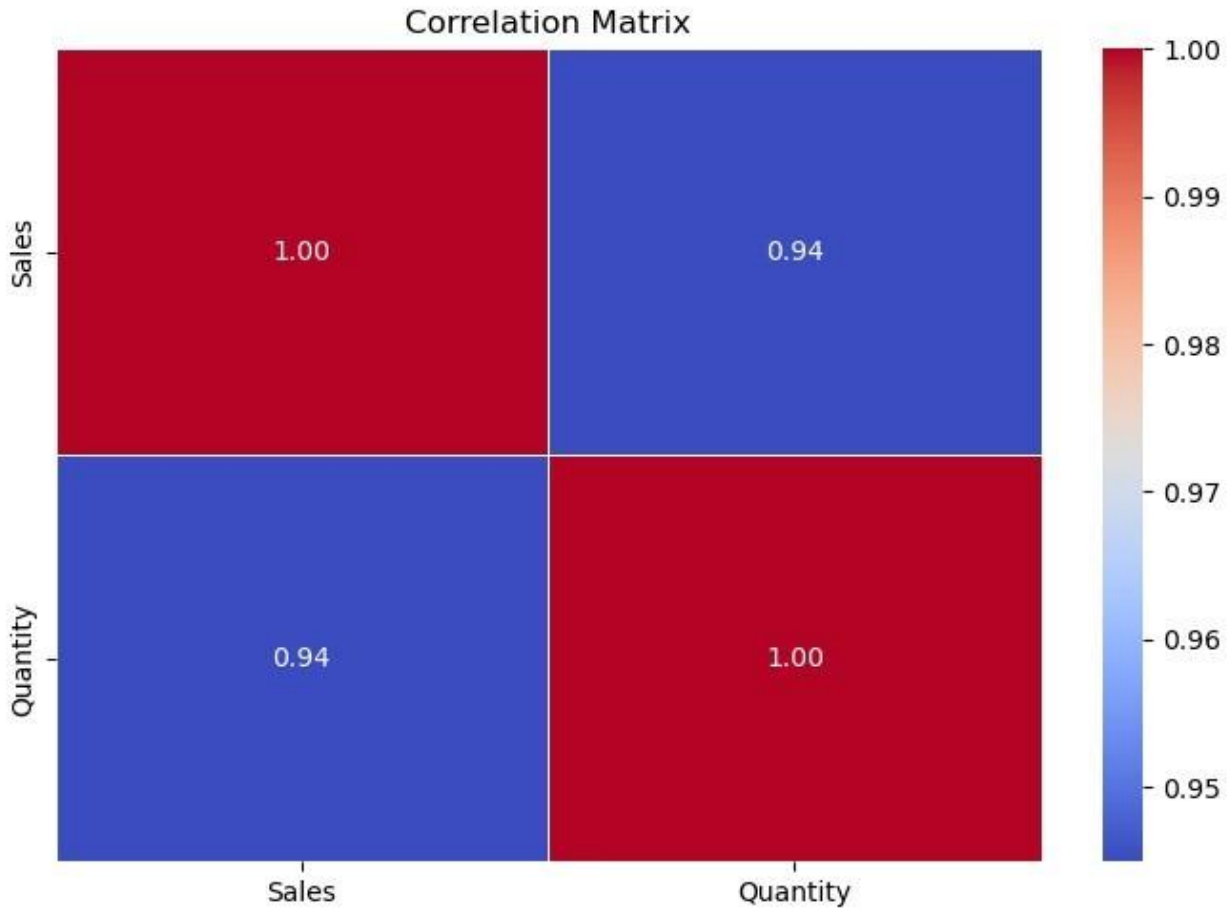
```
max    340.000000   8.000000
PRODUCT-WISE SALES AND QUANTITY:
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
CORRELATION MATRIX:

              Sales  Quantity
Sales      1.000000  0.944922
Quantity   0.944922  1.000000
```


Correlation Matrix
Sales
Quantity
1.00
0.94
0.94
1.00
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95